Polylogarithmic Approximation for Covering and Connecting Multi-Interface Networks

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Abstract

We consider the problem of covering/connecting a multi-interface network with a minimum cost assignment of interfaces to vertices. In the covering problem we demand that every edge is covered by at least one common interface assigned to its endpoints. In the connection problem we are given a set of terminals $U \subseteq V$ and we demand that the subgraph of activated edges spans all of them. We give an $O(\log m)$ approximation algorithm for the min-max cost criterion. The algorithm is based on rounding the optimal solution of a natural, set-cover-style linear programming relaxation of the problem. We also show how to use this LP to obtain a k-approximation algorithm, where k is the number of interfaces. Importantly, both of the aforementioned procedures work even when the cost of activating an interface depends on the interface as well as the vertex to which it is assigned. This improves the previously best known $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation, where b is a certain parameter of the input instance which can be as large as $\Omega(n)$ and $c_{\max}$ is the maximum cost of activating an interface and $O(k)$-approximation for the uniform cost case [Algorithmica 2012]. For the connection problem we devise an $O\left(\log^2 m\right)$-approximation algorithm, which is the first non-trivial approximation algorithm for this problem. The algorithm is based on a similar LP relaxation with additional cut constraints to ensure connectivity of the terminals.

2012 ACM Subject Classification Replace ccsdesc macro with valid one

Keywords and phrases Multi-Interface Networks, Approximation Algorithms, Covering, Connecting

Digital Object Identifier 10.4230/LIPIcs.CVIT.2016.23

1 Introduction

Designing Internet of Things (IoT) networks often demands establishing connection between large number of devices, which can be achieved by choosing between large variety of interfaces available at each device. Moreover, the devices present in the network may vary according to the interfaces they provide for the usage. For example some device may be able to connect by Bluetooth and Wi-Fi, while the others may provide Zigbee and Z-Wave interfaces. Since IoT devices are typically resource-constrained, it may be desirable to try to establish every relevant connection, while minimizing the amount of resources required to do so.

We model the above problem as one of two graph problems, depending on the type of the connection one wishes to establish. Each devices in the network is modeled as vertex of a graph. Additionally, with each such vertex we associate a set of labels representing the available interfaces. In the covering problem each edge in this graph represents a connection that we wish to establish by assigning a common interface to its endpoints. To see an example, see Figure 1. In the connection problem we are additionally given a set of terminals $U \subseteq V$, each edge can be activated by assigning a common interface to its endpoints and we require that the subgraph of activated edges connects all terminals (vertices outside U may be used as Steiner vertices). To see an example, see Figure 2. In both problems, we assume that activating an interface at a given vertex incurs some arbitrary cost. The goal is to minimize either the min-max cost (the maximum cost per vertex) or min-sum cost (sum of costs over all vertices) of the interface assignment. Hereby, we focus ourselved on the min-max cost criterion.

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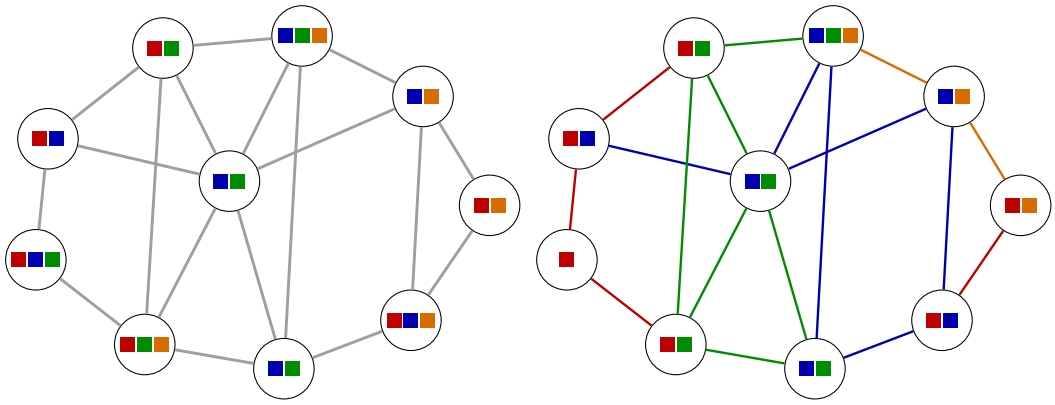
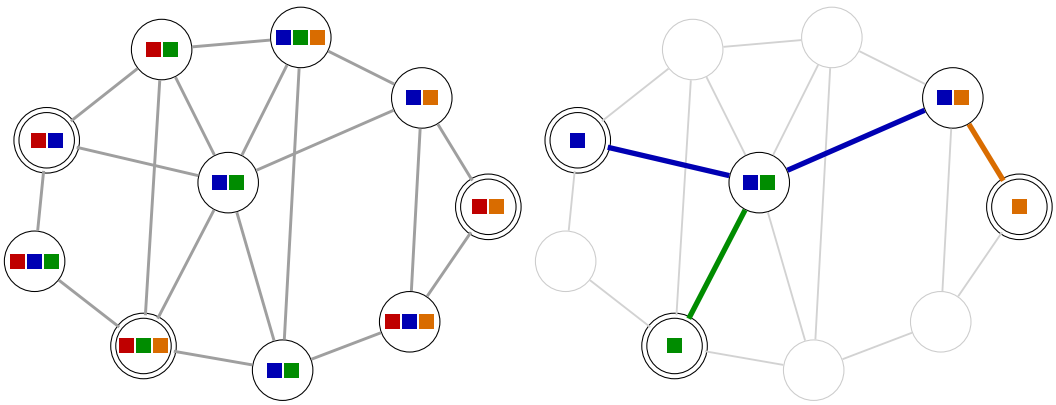
42nd Conference on Very Important Topics (CVIT 2016).
Editors: John Q. Open and Joan R. Access; Article No. 23; pp. 23:1–23:9
Leibniz International Proceedings in Informatics
LIPICS Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany



Available labels

Example covering assignment

Figure 1 An example instance of a multi-interface network with available interfaces at each vertex and a feasible covering assignment.



Available labels (double contour = terminal)

Example connecting assignment

Figure 2 An example instance of a multi-interface network with available interfaces at each vertex and a feasible connecting assignment.

1.1 Related Work

The algorithmic study of multi-interface networks goes back at least to the work of Klasing, Kosowski, and Navarra [12]. When k is bounded they show APX-hardness and provide an $\min\{\lceil \frac{k+1}{2}, \frac{2m}{n} \rceil\}$-approximation algorithm for the min-sum version of the covering problem with uniform costs. For unbounded k they give an $\min\{k-1, \sqrt{n}(1+\ln n)\}$-approximation for interface-dependent costs and prove that the problem is NP-hard to approximate within $O(\log k)$-factor. Assuming $U = V$, the min-sum connection problem for $k \geq 2$ is known to be APX-hard even for uniform costs and a $(2 - 1/k)$-approximation algorithm is known as well as 4/3-approximation for $k = 2$. Non uniform, interface dependent costs version admits 2-approximation [14]. The same research direction also led to results for other classical optimization problems in the multi-interface setting, including maximum matching [13] and flow optimization with bandwidth constraints [8].

The min-max coverage problem is known to be NP-hard for any fixed $\Delta \geq 5$ and $k \geq 16$ and cannot be approximated within a factor of $\eta \ln \Delta$ unless P=NP even for uniform costs on trees (for a certain constant η). On the algorithmic side the problem admits

an $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation algorithm, where b denotes the existence of a b-bounded ownership function as well as $O\{k \cdot c_{max}\}$ and $O\{\Delta \cdot c_{max}\}$-approximation algorithms. Assuming $U = V$, the min-max connection is known to be NP-hard for any fixed $\Delta \geq 3$ and $k \geq 10$, but is solvable for $k \leq 2$. By equivalence, for the case of trees the problem admits $O(\log n)$-approximation (since for trees $b = 1$) and $O\{k \cdot c_{max}\}$ and $O\{\Delta \cdot c_{max}\}$-approximation. No such results are known for the general case [7]

Another active line of research concerns constrained and parameterized versions of coverage. In the *p-coverage* setting, where each node may activate at most p interfaces, the problem is NP-complete already for $p = 2$ on graphs with maximum degree $\Delta \geq 4$, even with uniform costs [1]. Aloisio and Navarra showed that for $p = 2$ the problem is FPT with respect to treewidth tw via Courcelle's theorem. They also gave an explicit dynamic programming algorithms, for carvingwidth and pathwidth [1]. The pathwidth case was subsequently revisited and extended in [2]. Most recently the covering problem was studied in [4], establishing FPT results including an algorithm parametrized by $k + \mathrm{tw}$ and hardness lower bounds with respect to several structural parameters, including vertex cover number, feedback edge number and distance to clique.

1.2 Our contribution and techniques

In this paper we give a first $O(\log m)$-approximation algorithm for the min-max connection problem with non-uniform costs. This improves the previously known result of [7], which gave an $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation for the min-max covering version, where b is the parameter related to the existence of a b-owner function [9, 6]. This als improves the $O(\sqrt{n} \cdot \log n)$-approximation for the min-sum covering problem by [12]. Our approach is based on a set-cover-style LP relaxation together with a randomized rounding. We also show how to use the same LP relaxation to obtain a k-approximation algorithm for the min-sum and min-max covering problems, where k is the number of interfaces. All these guarantees hold in the non-uniform setting where activation costs depend on both the interface and the vertex it is assigned to.

2 Preliminaries

Let $G = (V, E)$ be a connected simple graph. Let $\lambda\colon V \to [k]$ be a function that assigns to each vertex $v \in V$ a set of available interfaces. In the *covering problem* we wish to find an assignment $\lambda_A\colon V \to [k]$ such that $\lambda_A(v) \subseteq \lambda(v)$ for all $v \in V$ and for every edge $uv \in E$ we have $\lambda_A(u) \cap \lambda_A(v) \neq \emptyset$. We call such an assignment a *covering assignment*. In the *connection problem* we are additionally given a set of *terminals* $U \subseteq V$. We wish to find an assignment $\lambda_A\colon V \to [k]$ such that $\lambda_A(v) \subseteq \lambda(v)$ for all $v \in V$ and the subgraph G_A of G induced by the edges $uv \in E$ with $\lambda_A(u) \cap \lambda_A(v) \neq \emptyset$ connects all terminals, i.e., every two vertices of U lie in the same connected component of G_A. Vertices in $V \setminus U$ may be used as Steiner vertices. We call such an assignment a *connecting assignment*. The special case $U = V$ corresponds to requiring G_A to be fully connected.

We additionally assume, that a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$ is given, which denotes the cost of assingining the interface $i \in [k]$ to vertex $v \in V$. The max cost of the assignment is defined as:

$$\mathtt{COST}_S = \max_{v \in V} \sum_{i \in \lambda_A(v)} c(i, v)\,.$$

The goal of the min-max assignment problem is to find an assignment λ_A that minimizes

$\texttt{COST}_S$. The sum cost of the assignment is defined as:

$$\texttt{COST}_M = \sum_{v \in V} \sum_{i \in \lambda_A(v)} c(i, v)\,.$$

The goal of the min-sum assignment problem is to find an assignment λ_A that minimizes $\texttt{COST}_M$. When not mentioned otherwise, we will focus on the min-max cost criterion.

Additionally, we need the following probability inequality known as the Chernoff bound [5]:

▶ **Theorem 1** (Chernoff Bound). *Let $X_1, \ldots, X_n$ be independent random variables taking values in $\{0, 1\}$ and let $X = \sum_{i=1}^{n} X_i$. Then for every $\delta > 0$:*

$$\Pr[X \geq (1+\delta) \cdot \mathbb{E}[X]] \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}} \right)^{\mathbb{E}[X]}$$

and for every $0 < \delta < 1$:

$$\Pr[X \leq (1-\delta) \cdot \mathbb{E}[X]] \leq \left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}} \right)^{\mathbb{E}[X]}.$$

3 LP based approximation algorithms

Below we show an $O(\log m)$ approximation algorithms for all four variants and $O(k)$-approximation for the covering problems. The idea of our ILP formulation and the algorithm is inspired by the rounding procedure known for set cover [15] and related problems [3]

3.1 Rounding the costs

Due to technical reasons, we show how to produce a polynomial number of instances, with cost function bounded by a polynomial in n. By testing consecutive powers of 2, we find the smallest number b, such that the costliest activated interface in the optimal solution has cost at most 2^b. For each vertex $v \in V$, we discard all of the interfaces from $\lambda(v)$ with cost larger than 2^b and we normalize the costs of remaining interfaces by dividing them by 2^b. Note that the number of guesses is polynomial. We also can assume that $\texttt{OPT} \geq 1/2$, since the cost of any solution is at least the cost of activating one interface, which is at least $1/2$ by our rounding.

3.2 ILP formulation of covering multi-interface networks

We start with the following ILP formulation of the min-sum covering assignment problem. Let $x_{i,v}$ be a binary variable that is 1 if and only if the interface $i \in \lambda(v)$ is assigned to vertex $v \in V$. We consider two types of vertices. For a given vertex $v \in V$, if $\sum_{i \in \lambda(v)} c(i, v) \leq 1$, then we can safely assume that all interfaces in $\lambda(v)$ are activated at v, since the cost of activating all of them is at most 1 and thus at most $2 \cdot \texttt{OPT}$. We call such vertices *cheap*. For the remaining vertices, we call them *expensive*. Note that for every expensive vertex v, we can assume that $\sum_{i \in \lambda(v)} x_{i,v} \cdot c(i, v) \geq 1$, since this will always result in a solution of cost at most $2 \cdot \texttt{OPT}$. Thus, we can write the following ILP formulation of the min-max covering

assignment problem:

$$\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \leq M \quad \forall v \in V \\
& \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1 \quad \forall uv \in E \\
& \begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} < 1 \\ \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} \geq 1 & \text{otherwise} \end{cases} \quad \forall v \in V \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v)
\end{aligned}$$

Note that the above formulation is not linear due to the presence of the min function. However, by adding an additional variable $y_{i,uv}$ we can linearize the constraints of type $\sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1$ by replacing them with $\sum_{i \in \lambda(u) \cap \lambda(v)} y_{i,uv} \geq 1$ together with the constraints $y_{i,uv} \leq x_{i,u}$ and $y_{i,uv} \leq x_{i,v}$ for every $i \in \lambda(u) \cap \lambda(v)$ and $uv \in E$. Thus, we can solve the LP relaxation of this formulations in polynomial time.

3.3 Randomized $O(\log m)$-approximation algorithm for the covering problem

Algorithm 1 A relaxation-based $O(\log m)$-approximation algorithm.

1. Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}$.
2. Pick independent uniformly random thresholds $t_i \in (0,1)$, for each $i \in [k]$.
3. Set $\lambda_A(v) = \{i \in \lambda(v) \colon 3 \ln m \cdot x_{i,v} \geq t_i\}$, for every $v \in V$.
4. **return** $\{\lambda_A(v)\}_{v \in V}$.

▶ **Theorem 2.** *The above algorithm is an $O(\log m)$-approximation algorithm for the min-max covering assignment problems which succeeds with high probability.*

Proof. We have the following lemmas:

▶ **Lemma 3.** *The algorithm returns a solution of cost at most $9 \ln m \cdot \mathtt{OPT}$ with high probability.*

Proof. Fix a vertex $v \in V$. We want to show that with high probability, the cost of activating interfaces at v is at most $9 \ln m$ times the cost of activating interfaces at v in the fractional solution. If v is cheap, then the cost of activating all interfaces at v is at most 1, so the claim holds. Assume that v is expensive. By $X_{v,i}$ denote the random variable that is 1 if $3 \ln m \cdot x_{i,v} \geq t_i$ and 0 otherwise. Then we can write $c(\lambda_A(v)) = \sum_{i \in \lambda(v)} X_{v,i} \cdot c(i,v)$. We have: $\mathbb{E}[X_{v,i}] = \Pr[i \in \lambda_A(v)] = 3 \ln m \cdot x_{i,v}$. Thus, $\mathbb{E}[c(\lambda_A(v))] = 3 \ln m \cdot \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v)$. By applying the Chernoff bound with $\delta = 2$ we get that

$$\Pr[c(\lambda_A(v)) \geq 3 \cdot \mathbb{E}[c(\lambda_A(v))]] \leq \left(\frac{e^2}{3^3}\right)^{\mathbb{E}[c(\lambda_A(v))]} \leq \exp\{-\mathbb{E}[c(\lambda_A(v))]\} \leq \frac{1}{m^3}$$

where the last inequality is by using the fact that, $\sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \geq 1$. By applying the union bound over all n vertices we get that with probability at least $1 - \frac{n}{m^3} \geq 1 - \frac{1}{m}$ (assuming $m \geq 2$) for every $v \in V$, $c(\lambda_A(v)) \leq 9 \ln m \cdot \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v)$. Thus, with high probability we have $\mathtt{COST}(\lambda_A) \leq 9 \ln m \cdot \max_{v \in V} \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) = 9 \ln m \cdot \mathtt{OPT}$. ◀

▶ **Lemma 4.** *With high probability, the above algorithm returns a covering assignment.*

Proof. Consider an edge $uv \in E$. We have:

$$\begin{aligned}
\Pr[\lambda_A(u) \cap \lambda_A(v) = \emptyset] &= \prod_{i \in \lambda(u) \cap \lambda(v)} \Pr[t_i > 3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}] \\
&= \prod_{i \in \lambda(u) \cap \lambda(v)} \{1 - 3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&\leq \prod_{i \in \lambda(u) \cap \lambda(v)} \exp\{-3 \ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&= \exp\left\{-3 \ln m \cdot \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\}\right\} \\
&\leq \exp\{-3 \ln m\} = \frac{1}{m^3}
\end{aligned}$$

By applying the union bound over all m edges we get that the algorithm returns a covering assignment with probability at least $1 - \frac{m}{m^3} = 1 - \frac{1}{m^2}$. ◀

Combining the above lemmas gives the claim. ◀

3.4 A k-approximation algorithm for the covering problem

Algorithm 2 A relaxation-based k-approximation algorithm.

1. Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}$.
2. Set $\lambda_A(v) = \{i \in \lambda(v) \colon x_{i,v} \geq \frac{1}{k}\}$, for every $v \in V$.
3. **return** $\{\lambda_A(v)\}_{v \in V}$.

▶ **Theorem 5.** *The above algorithm is a k-approximation algorithm for both the min-sum and the min-max covering assignment problems.*

Proof. Consider an edge $uv \in E$. By averaging, there exists an interface $i \in \lambda(u) \cap \lambda(v)$ such that $\min\{x_{i,u}, x_{i,v}\} \geq \frac{1}{k}$. Thus, $i \in \lambda_A(u) \cap \lambda_A(v)$ so λ_A is a covering assignment. Moreover, since $x_{i,v} \geq \frac{1}{k}$ for every $i \in \lambda_A(v)$, we have that $c(\lambda_A(v)) = \sum_{i \in \lambda_A(v)} c(i, v) \leq k \cdot \sum_{i \in \lambda_A(v)} x_{i,v} \cdot c(i, v)$. Thus, depending on the cost function, we have $\texttt{COST}(\lambda_A) \leq k \cdot \max_{v \in V} \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i, v) = k \cdot \texttt{OPT}$. ◀

3.5 ILP formulation for the connection problem

In this section we again assume that $\texttt{OPT} \geq 1/2$, the cost of activating any interface is at most 1 and we distinguish between cheap and expensive vertices as before. We use a similar LP to model the connection problem, however we use additional flow variables to encode the connectivity requirement. The $x_{i,v}$ variables are defined as before. For each edge e we add an additional flow variable y_e which is 1 if and only if the edge e is covered by the solution. Connectivity of the terminals U is enforced via cut constraints: we require that for every cut $(S, V \setminus S)$ that separates at least one pair of terminals, the total y-weight of edges crossing

the cut is at least 1. Then we can formulate the min-max connection problem as follows:

$$
\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \leq M \quad \forall v \in V \\
& \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq y_{uv} \quad \forall uv \in E \\
& \begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} < 1 \\ \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} \geq 1 & \text{otherwise} \end{cases} \quad \forall v \in V \\
& y(\delta(S)) \geq 1 \quad \forall S \subset V \colon S \cap U \neq \emptyset,\ (V \setminus S) \cap U \neq \emptyset \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) \\
& y_{uv} \in \{0,1\} \quad \forall uv \in E
\end{aligned}
$$

Observe that the above ILP has exponential number of constraints, but we can solve its LP relaxation in polynomial time by using the ellipsoid method and a separation oracle for the cut constraints, which can be implemented using a max-flow algorithm.

3.6 Randomized $O\left(\log^2 m\right)$-approximation algorithm for the connection problem

Algorithm 3 A relaxation-based $O(\log^2 m)$-approximation algorithm.

1. Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}, \{y_{uv}\}_{uv \in E}$.
2. For every edge $e \in E$, let $p_e = \min\{1, 3 \ln m \cdot y_e\}$. Pick independent uniformly random variables Y_e such that $Y_e = 1$ with probability p_e and $Y_e = 0$ otherwise. Let H be the subgraph of G induced by the edges e such that $Y_e = 1$.
3. Use the rounding algorithm for the covering problem described on the LP solution restricted to the subgraph H to obtain a covering assignment λ_A for H.
4. **return** $\{\lambda_A(v)\}_{v \in V}$.

▶ **Theorem 6.** *There exists an $O\left(\log^2 m\right)$-approximation algorithm for both the min-sum and the min-max connection assignment problems which succeeds with high probability.*

Proof. We show that the algorithm for the covering problem described in the previous section can be used to obtain an $O\left(\log^2 m\right)$-approximation algorithm for the connection problem. We start with the following lemma:

▶ **Lemma 7.** *With high probability, the subgraph H connects all terminals in U.*

Proof. Fix any $S \subset V$ with $S \cap U \neq \emptyset$ and $(V \setminus S) \cap U \neq \emptyset$. Let $\ell \in \mathbb{N}$, such that $\ell \leq y(\delta(S)) < \ell + 1$. We have:

$$
\begin{aligned}
\Pr[|\delta_H(S)| = 0] &= \prod_{e \in \delta(S)} \Pr[Y_e = 0] = \prod_{e \in \delta(S)} (1 - p_e) \\
&\leq \prod_{e \in \delta(S)} \exp\{-p_e\} \leq \exp\left\{ - \sum_{e \in \delta(S)} \min\{1,\, 3 \ln m \cdot y_e\} \right\} \\
&\leq \exp\{-3\ell \ln m\} = \frac{1}{m^{3\ell}}
\end{aligned}
$$

where the last inequality uses the LP constraint $y(\delta(S)) \geq 1$ for every U-separating cut, so $\sum_{e \in \delta(S)} y_e \geq \ell$.

A cut in G is called an α-minimum cut if its value is less than α times the value of a minimum cut in G. We now make a use of the following Karger's lemma [11, 10]:

▶ **Lemma 8** (Karger's lemma). *The number of α-minimum cuts in a graph is at most $n^{2\alpha}$.*

Observe that by construction of the LP, every U-separating fractional cut has a value of at least 1, so there are at most $n^{2\ell}$ U-separating cuts of value less than ℓ. By applying the union bound over all U-separating cuts we get that:

$$\Pr[\exists\, U\text{-sep. cut } S\colon |\delta_H(S)| = 0] \leq \sum_{\ell=1}^{\infty} \sum_{\substack{S \subset V,\, S \cap U \neq \emptyset \\ (V \setminus S) \cap U \neq \emptyset,\, y(\delta(S)) < \ell+1}} \Pr[|\delta_H(S)| = 0]$$

$$\leq \sum_{\ell=1}^{\infty} n^{2\ell} \cdot \frac{1}{m^{3\ell}} = \sum_{\ell=1}^{\infty} \frac{1}{m^{\ell}} = \frac{1}{m-1}$$

Thus, with high probability, no U-separating cut is empty in H, i.e., H spans U. ◀

Now we want to show that by covering H we can obtain good approximation for connecting G. We have the following lemma:

▶ **Lemma 9.** *With high probability, the cost of covering H is at most $9 \ln m \cdot \mathtt{OPT}$.*

Proof. Let $\bar{\mathbf{x}}, \bar{\mathbf{y}}$ be the fractional solution such that every variable is scaled up by a factor of $3 \ln m$. Clearly, this is a valid LP solution. Moreover, by the linearity of the LP objective function, we know that the cost of $\bar{\mathbf{x}}, \bar{\mathbf{y}}$ is at most $3 \ln m \cdot \mathtt{OPT}$. Since for every edge $e \in E$, we have that $\Pr[Y_e = 1] = \min\{1, 3 \ln m \cdot y_e\}$, by linearity of expectations we know that the expected cost of covering H, call it $\mathtt{COST}(H)$ is at most $3 \ln m \cdot \mathtt{OPT}$. By applying the Chernoff bound with $\delta = 2$ we have that:

$$\Pr[\mathtt{COST}(H) \geq 3 \cdot \mathbb{E}[\mathtt{COST}(H)]] \leq \left(\frac{e^2}{3^3}\right)^{\mathbb{E}[\mathtt{COST}(H)]} \leq \exp\{-\mathbb{E}[\mathtt{COST}(H)]\} \leq \frac{1}{m^3}.$$

where in the last inequality we used the fact that due to our treatment of cheap and expensive vertices we get that $\mathbb{E}[\mathtt{COST}(H)] \geq 1$. Thus, with high probability, the cost of covering H is at most $9 \ln m \cdot \mathtt{OPT}$. ◀

Observe that the any LP solution restricted to a subgraph of G has cost at most the cost of the original solution. Since we know that the rounding algorithm for the covering problem returns a covering assignment of cost at most $9 \ln m \cdot \mathtt{OPT}_{LP}$ with high probability, where $\mathtt{OPT}_{LP}$ is the cost of the LP solution, we get that with high probability the cost of the returned connecting assignment is at most $O\big(\log^2 m\big) \cdot \mathtt{OPT}_{LP} \leq O\big(\log^2 m\big) \cdot \mathtt{OPT}$. ◀

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